numpy project 1

In [1]:
```
import numpy as np
```

In [3]:
```
ones_arr = np.ones((5,5))
```

In [4]:
```
ones_arr
```

Out[4]:
```
array([[1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.]])
```

In [5]:
```
ones_arr = np.ones((5,5),dtype=int)
```

In [6]:
```
ones_arr
```

Out[6]:
```
array([[1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1]])
```

In [7]:
```
zeros_arr = np.zeros((3,3),dtype=int)
zeros_arr
```

Out[7]:
```
array([[0, 0, 0],
       [0, 0, 0],
       [0, 0, 0]])
```

In [8]:
```
ones_arr
```

Out[8]:
```
array([[1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1]])
```

In [9]:
```
ones_arr*255
```

Out[9]:
```
array([[255, 255, 255, 255, 255],
       [255, 255, 255, 255, 255],
       [255, 255, 255, 255, 255],
       [255, 255, 255, 255, 255],
       [255, 255, 255, 255, 255]])
```

In [10]:
```
zeros_arr
```

Out[10]:
```
array([[0, 0, 0],
       [0, 0, 0],
       [0, 0, 0]])
```

In [11]:
```
ones_arr
```

Out[11]:
```
array([[1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1]])
```

In [12]:
```
import matplotlib.pyplot as plt
```

matplotlib is the library for visualisation # numpy for ND array

In [15]:
```
%matplotlib inline
```

the above code will keep all the pictures inside the box or graph

In [16]:
```
from PIL import Image
```

PIL represents python imaging library

In [18]:
```
king_img = Image.open(r"C:\Users\Abhilash Reddy\Desktop\king.jpg")
```

In [19]:
```
king_img
```

Out[19]:

In [20]:
```
type(king_img)
```

Out[20]:
```
PIL.JpegImagePlugin.JpegImageFile
```

In [22]:
```
king_arr = np.asarray(king_img)
```

In [23]:
```
king_arr
```

Out[23]:
```
array([[[195, 211, 224],
        [193, 209, 222],
        [191, 207, 220],
        ...,
        [199, 211, 235],
        [199, 211, 235],
        [199, 211, 235]],

       [[200, 216, 229],
        [198, 214, 227],
        [194, 210, 223],
        ...,
        [195, 207, 231],
        [195, 207, 231],
        [195, 207, 231]],

       [[205, 222, 232],
        [203, 220, 230],
        [199, 216, 226],
        ...,
        [190, 202, 226],
        [190, 202, 226],
        [190, 202, 226]],

       ...,

       [[ 22,  35,   7],
        [ 18,  31,   3],
        [ 17,  30,   2],
        ...,
        [  1,  11,   2],
        [  0,   4,   0],
        [  9,  19,  10]],

       [[ 20,  35,   4],
        [ 17,  32,   3],
        [ 15,  30,   1],
        ...,
        [  0,   7,   0],
        [ 18,  28,  19],
        [131, 141, 132]],

       [[ 17,  32,   1],
        [ 16,  31,   0],
        [ 15,  30,   1],
        ...,
        [  7,  17,   8],
        [130, 140, 131],
        [122, 132, 123]]], shape=(685, 1200, 3), dtype=uint8)
```

In [24]:
```
type(king_arr)
```

Out[24]:
```
numpy.ndarray
```

In [25]:
```
king_arr.shape
```

Out[25]: (685, 1200, 3)

In [26]:
```
plt.imshow(king_arr)
```

Out[26]: <matplotlib.image.AxesImage at 0x1c3631e2900>

In [27]:
```
king_red = king_arr.copy()
```

In [28]:
```
king_red
```

Out[28]:
```
array([[[195, 211, 224],
        [193, 209, 222],
        [191, 207, 220],
        ...,
        [199, 211, 235],
        [199, 211, 235],
        [199, 211, 235]],

       [[200, 216, 229],
        [198, 214, 227],
        [194, 210, 223],
        ...,
        [195, 207, 231],
        [195, 207, 231],
        [195, 207, 231]],

       [[205, 222, 232],
        [203, 220, 230],
        [199, 216, 226],
        ...,
        [190, 202, 226],
        [190, 202, 226],
        [190, 202, 226]],

       ...,

       [[ 22,  35,   7],
        [ 18,  31,   3],
        [ 17,  30,   2],
        ...,
        [  1,  11,   2],
        [  0,   4,   0],
        [  9,  19,  10]],

       [[ 20,  35,   4],
        [ 17,  32,   3],
        [ 15,  30,   1],
        ...,
        [  0,   7,   0],
        [ 18,  28,  19],
        [131, 141, 132]],

       [[ 17,  32,   1],
        [ 16,  31,   0],
        [ 15,  30,   1],
        ...,
        [  7,  17,   8],
        [130, 140, 131],
        [122, 132, 123]]], shape=(685, 1200, 3), dtype=uint8)
```

In [30]:
```
king_arr == king_red # to check wheter it is right or wrong
```

Out[30]:
```
array([[[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       ...,

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]]], shape=(685, 1200, 3))
```

In [31]:
```
plt.imshow(king_red)
```

Out[31]:
```
<matplotlib.image.AxesImage at 0x1c3635d4f50>
```

In [32]:
```
king_red.shape
```

Out[32]: (685, 1200, 3)

now we make all the rows and columns as zero

In [33]:
```
plt.imshow(king_red[:,:,0])
```

Out[33]: <matplotlib.image.AxesImage at 0x1c3635fead0>

In [34]:
```
king_red[:,:,0]
```

Out[34]:
```
array([[195, 193, 191, ..., 199, 199, 199],
       [200, 198, 194, ..., 195, 195, 195],
       [205, 203, 199, ..., 190, 190, 190],
       ...,
       [ 22,  18,  17, ...,   1,   0,   9],
       [ 20,  17,  15, ...,   0,  18, 131],
       [ 17,  16,  15, ...,   7, 130, 122]],
      shape=(685, 1200), dtype=uint8)
```

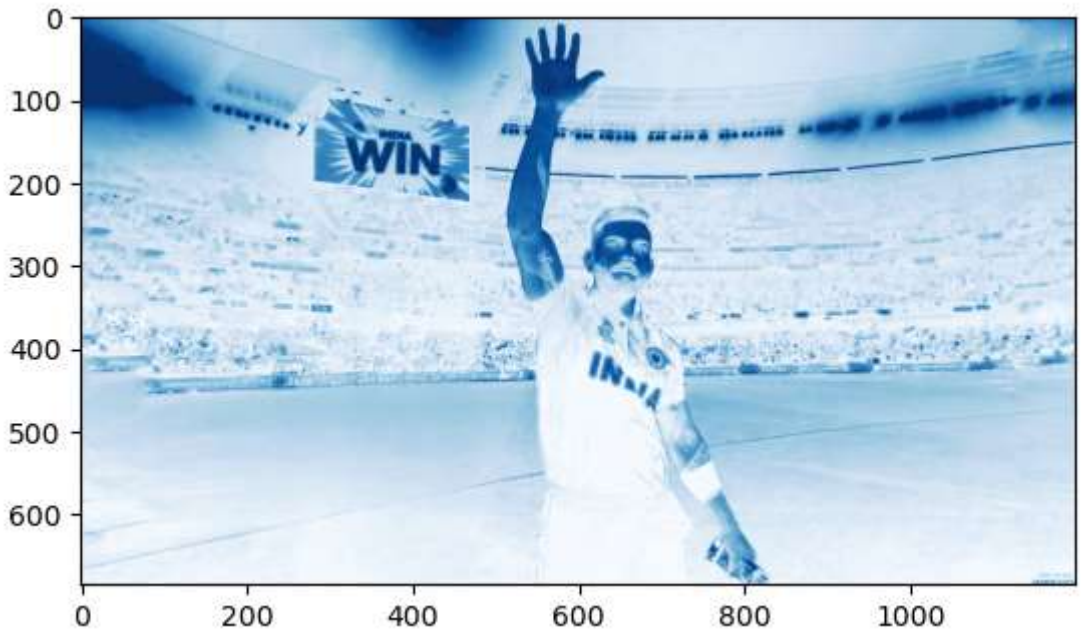
google = matplotib documentation colormap reference cmap = the above google search colours

In [36]:
```
plt.imshow(king_red[:,:,0], cmap='gray' ) # to get gray colour
```

Out[36]:
```
<matplotlib.image.AxesImage at 0x1c364697750>
```

In [41]:
```
plt.imshow(king_red[:,:,0], cmap='Blues' )
```

Out[41]:
```
<matplotlib.image.AxesImage at 0x1c3673a2fd0>
```

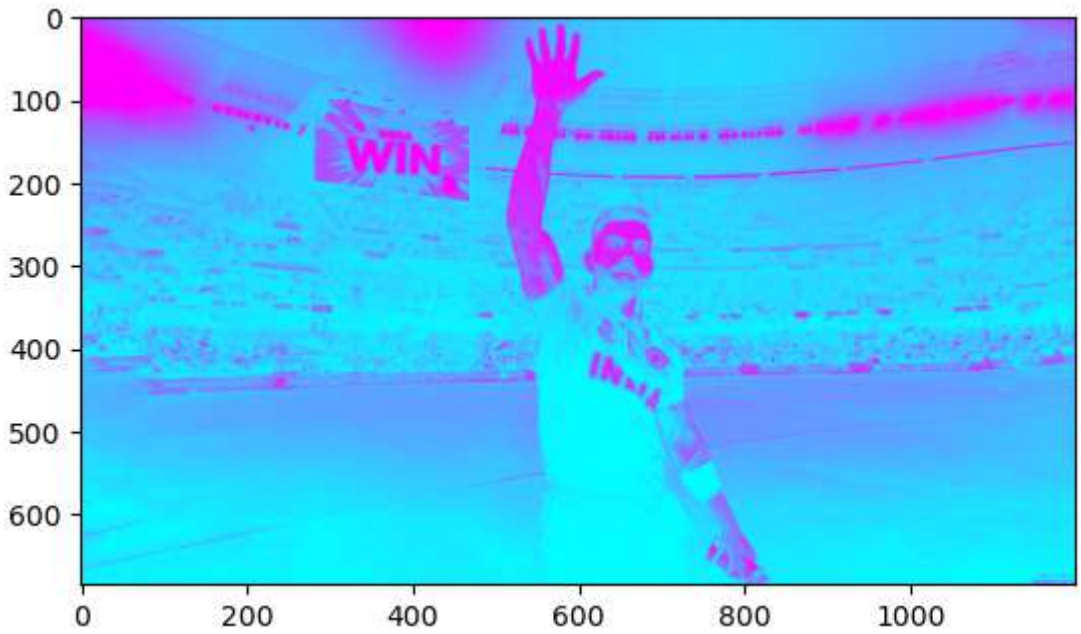


In [42]:
```
plt.imshow(king_red[:,:,0], cmap='magma' )
```

Out[42]:
```
<matplotlib.image.AxesImage at 0x1c367415090>
```

In [43]:
```
plt.imshow(king_red[:,:,0], cmap='cool' )
```

Out[43]:
```
<matplotlib.image.AxesImage at 0x1c367433610>
```



In [45]:
```
plt.imshow(king_red[:,:,0], cmap='Reds' )
```

Out[45]:
```
<matplotlib.image.AxesImage at 0x1c36759e210>
```

In [47]:
```
plt.imshow(king_red[:,:,0], cmap='Greys' )
```

Out[47]:
```
<matplotlib.image.AxesImage at 0x1c3676d0910>
```

In [48]:
```
plt.imshow(king_red[:,:,1], cmap='gray' )
```

Out[48]:
```
<matplotlib.image.AxesImage at 0x1c367866e90>
```

In [53]:
```
plt.imshow(king_red[:,:,2], cmap='gray' )
```

Out[53]:
```
<matplotlib.image.AxesImage at 0x1c3679174d0>
```

In [59]:
```
plt.imshow(king_red[:,:,3], cmap='grey' )
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[59], line 1
----> 1 plt.imshow(king_red[:,:,3], cmap='grey' )

IndexError: index 3 is out of bounds for axis 2 with size 3
```

In [56]:
```
king_red[:,:,0]
```

Out[56]:
```
array([[195, 193, 191, ..., 199, 199, 199],
       [200, 198, 194, ..., 195, 195, 195],
       [205, 203, 199, ..., 190, 190, 190],
       ...,
       [ 22,  18,  17, ...,   1,   0,   9],
       [ 20,  17,  15, ...,   0,  18, 131],
       [ 17,  16,  15, ...,   7, 130, 122]],
      shape=(685, 1200), dtype=uint8)
```

In [57]:
```
king_red[:,:,1]
```

Out[57]:
```
array([[211, 209, 207, ..., 211, 211, 211],
       [216, 214, 210, ..., 207, 207, 207],
       [222, 220, 216, ..., 202, 202, 202],
       ...,
       [ 35,  31,  30, ...,  11,   4,  19],
       [ 35,  32,  30, ...,   7,  28, 141],
       [ 32,  31,  30, ...,  17, 140, 132]],
      shape=(685, 1200), dtype=uint8)
```

In [58]:
```
king_red[:,:,2]
```

Out[58]:
```
array([[224, 222, 220, ..., 235, 235, 235],
       [229, 227, 223, ..., 231, 231, 231],
       [232, 230, 226, ..., 226, 226, 226],
       ...,
       [  7,   3,   2, ...,   2,   0,  10],
       [  4,   3,   1, ...,   0,  19, 132],
       [  1,   0,   1, ...,   8, 131, 123]],
      shape=(685, 1200), dtype=uint8)
```

king_red[:,:,1] lets make every array to zero

In [60]:
```
king_red[:,:,1] = 0
```

In [61]:
```
king_red[:,:,1]
```

Out[61]:
```
array([[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       ...,
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0]], shape=(685, 1200), dtype=uint8)
```

In [63]:
```
plt.imshow(king_red)
```

Out[63]:
```
<matplotlib.image.AxesImage at 0x1c3679d8050>
```

In [64]:
```
king_red[:,:,2]
```

Out[64]:
```
array([[224, 222, 220, ..., 235, 235, 235],
       [229, 227, 223, ..., 231, 231, 231],
       [232, 230, 226, ..., 226, 226, 226],
       ...,
       [  7,   3,   2, ...,   2,   0,  10],
       [  4,   3,   1, ...,   0,  19, 132],
       [  1,   0,   1, ...,   8, 131, 123]],
      shape=(685, 1200), dtype=uint8)
```

In [65]:
```
king_red[:,:,2] = 0
```

In [66]:
```
king_red[:,:,2]
```

Out[66]:
```
array([[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       ...,
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0]], shape=(685, 1200), dtype=uint8)
```

In [67]:
```
plt.imshow(king_red)
```

Out[67]:
```
<matplotlib.image.AxesImage at 0x1c36796a5d0>
```

In [68]:
```
king_arr
```

Out[68]:
```
array([[[195, 211, 224],
        [193, 209, 222],
        [191, 207, 220],
        ...,
        [199, 211, 235],
        [199, 211, 235],
        [199, 211, 235]],

       [[200, 216, 229],
        [198, 214, 227],
        [194, 210, 223],
        ...,
        [195, 207, 231],
        [195, 207, 231],
        [195, 207, 231]],

       [[205, 222, 232],
        [203, 220, 230],
        [199, 216, 226],
        ...,
        [190, 202, 226],
        [190, 202, 226],
        [190, 202, 226]],

       ...,

       [[ 22,  35,   7],
        [ 18,  31,   3],
        [ 17,  30,   2],
        ...,
        [  1,  11,   2],
        [  0,   4,   0],
        [  9,  19,  10]],

       [[ 20,  35,   4],
        [ 17,  32,   3],
        [ 15,  30,   1],
        ...,
        [  0,   7,   0],
        [ 18,  28,  19],
        [131, 141, 132]],

       [[ 17,  32,   1],
        [ 16,  31,   0],
        [ 15,  30,   1],
        ...,
        [  7,  17,   8],
        [130, 140, 131],
        [122, 132, 123]]], shape=(685, 1200, 3), dtype=uint8)
```

In [69]:
```
king_red
```

Out[69]:
```
array([[[195,   0,   0],
        [193,   0,   0],
        [191,   0,   0],
        ...,
        [199,   0,   0],
        [199,   0,   0],
        [199,   0,   0]],

       [[200,   0,   0],
        [198,   0,   0],
        [194,   0,   0],
        ...,
        [195,   0,   0],
        [195,   0,   0],
        [195,   0,   0]],

       [[205,   0,   0],
        [203,   0,   0],
        [199,   0,   0],
        ...,
        [190,   0,   0],
        [190,   0,   0],
        [190,   0,   0]],

       ...,

       [[ 22,   0,   0],
        [ 18,   0,   0],
        [ 17,   0,   0],
        ...,
        [  1,   0,   0],
        [  0,   0,   0],
        [  9,   0,   0]],

       [[ 20,   0,   0],
        [ 17,   0,   0],
        [ 15,   0,   0],
        ...,
        [  0,   0,   0],
        [ 18,   0,   0],
        [131,   0,   0]],

       [[ 17,   0,   0],
        [ 16,   0,   0],
        [ 15,   0,   0],
        ...,
        [  7,   0,   0],
        [130,   0,   0],
        [122,   0,   0]]], shape=(685, 1200, 3), dtype=uint8)
```

In [70]:
```
king_img
```

Out[70]:

In [72]:
```
arr1 = np.asarray(king_img)
```

In [73]:
```
type(arr1)
```

Out[73]:
```
numpy.ndarray
```

In [74]:
```
arr1.shape
```

Out[74]:
```
(685, 1200, 3)
```

In [75]:
```
plt.imshow(arr1)
```

Out[75]:
```
<matplotlib.image.AxesImage at 0x1c367a48b90>
```

gen ai projects needs these imaging concept

In [76]:
```
horse_img1 = arr1.copy()
```

In [78]:
```
horse_img1[:,:,0] = 0 # we make every row and column to zero
```

In [79]:
```
plt.imshow(horse_img1)
```

Out[79]:
```
<matplotlib.image.AxesImage at 0x1c367d0b110>
```

In [80]:
```
horse_img1[:,:,1]
```

Out[80]:
```
array([[211, 209, 207, ..., 211, 211, 211],
       [216, 214, 210, ..., 207, 207, 207],
       [222, 220, 216, ..., 202, 202, 202],
       ...,
       [ 35,  31,  30, ...,  11,   4,  19],
       [ 35,  32,  30, ...,   7,  28, 141],
       [ 32,  31,  30, ...,  17, 140, 132]],
      shape=(685, 1200), dtype=uint8)
```

In [83]:
```
horse_img1[:,:,1] = 0 # we make all the values into 1
```

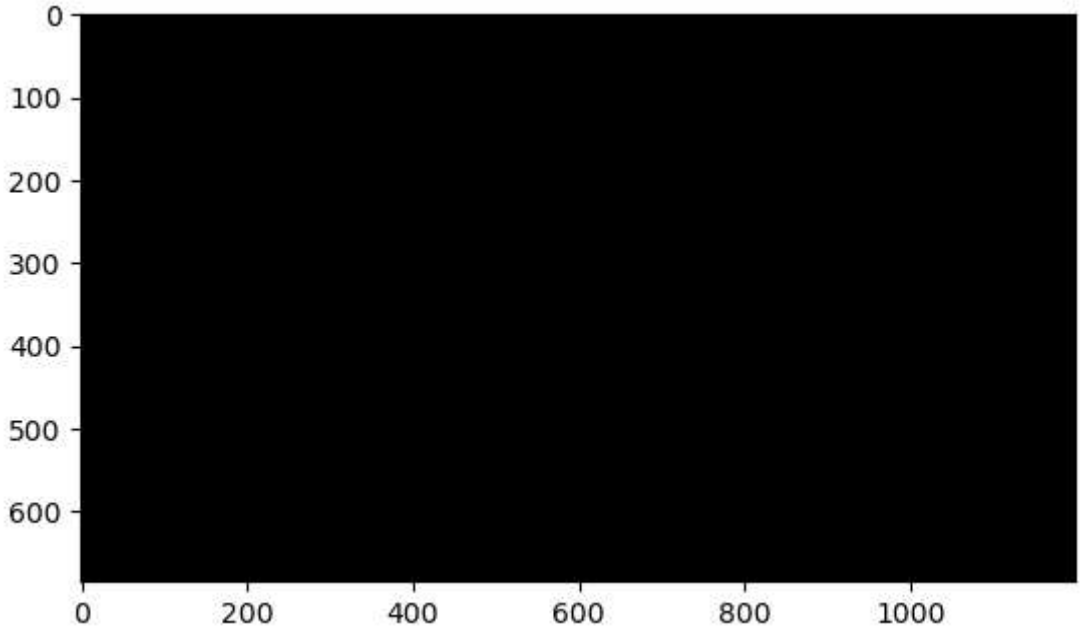
In [82]:
```
plt.imshow(horse_img1)
```

Out[82]:
```
<matplotlib.image.AxesImage at 0x1c367d796d0>
```

In [90]:
```
horse_img1[:,:,2] = 0 # converting all the values into 2
```

In [85]:
```
plt.imshow(horse_img1)
```

Out[85]:
```
<matplotlib.image.AxesImage at 0x1c369f7fc50>
```



In [89]:
```
horse_img1[:,:,3] = 0 # we can only use 0,1,2,
```

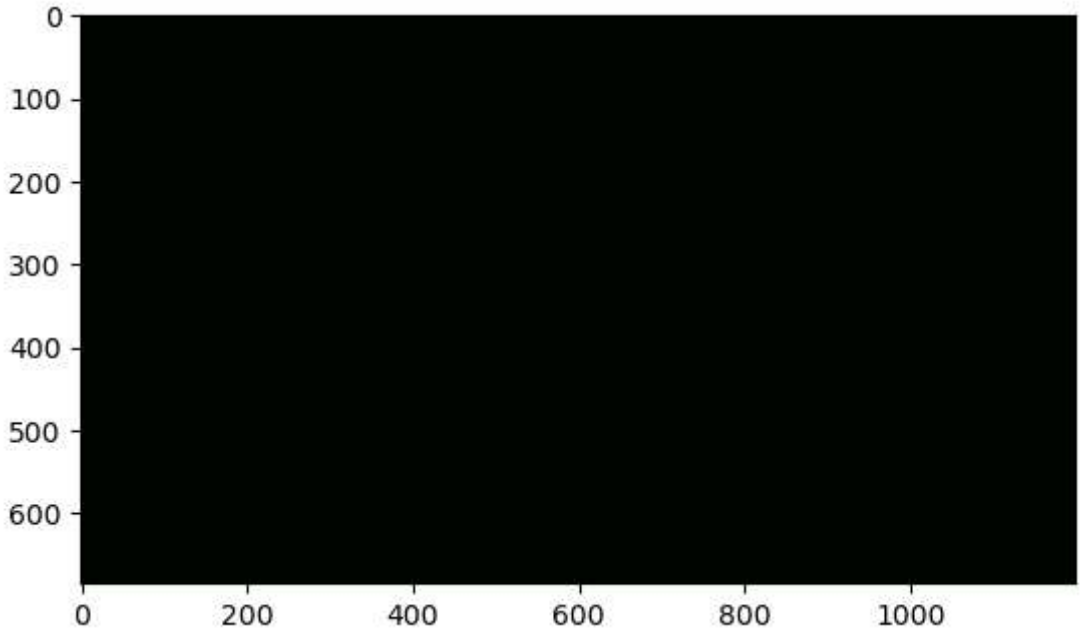
```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[89], line 1
----> 1 horse_img1[:,:,3] = 0 # we can only use 0,1,2,

IndexError: index 3 is out of bounds for axis 2 with size 3
```

In [91]:
```
horse_img1[:,:,1] = 5
```

In [92]:
```
plt.imshow(horse_img1)
```

Out[92]:
```
<matplotlib.image.AxesImage at 0x1c369e82210>
```



In [ ]: